

PSO-VNS

Phase 1: PSO

swarm reaches the
feasible region
 ωB evaluations

↓ best layout

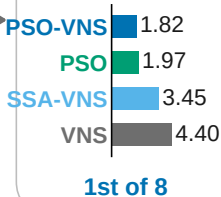
Phase 2: VNS

compass search and
shaking refine it
 $(1-\omega)B$ evaluations

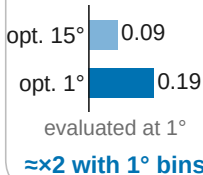
Controlled evaluation

2 wind data sets $\times$ 3 farm sizes
Horns Rev 1 · Lillgrund · IEA37

Average rank



Gain vs PSO (pp)



Findings

- ✓ ranks first of 8 methods
 - ✓ +0.19 pp over PSO with 1° bins
 - ✓ feasible 30/30 vs PSO 19/30*
 - GA, SLSQP lead in 2 settings
- *Horns Rev 1